

Refutation of conjecture 00000000471

The hypertree count of $K_n^{(3)}$: a closed form that exceeds the number of subhypergraphs

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Abstract

Conjecture 00000000471 proposes the closed form $t(K_n^{(3)}) = n^{\binom{n-1}{2}-1} \prod_{i=1}^{n-1} (i^2 - i + 1)$ for the number of hypertrees of the complete 3-uniform hypergraph, and states that it covers all known values for $n \leq 6$. We give two independent refutations. First, a hypertree is a subhypergraph and is therefore determined by a subset of the $\binom{n}{3}$ edges, so $t(K_n^{(3)}) \leq 2^{\binom{n}{3}}$; the conjectured formula violates this bound at every n with $3 \leq n \leq 13$, and in particular at all six values the conjecture claims to cover. This argument uses nothing about connectedness or minimality. Second, exhaustive enumeration gives $t(K_3^{(3)}) = 1$, $t(K_4^{(3)}) = 6$ and $t(K_5^{(3)}) = 25$, against 3, 336 and 853125 from the formula. Both refutations are formalised in core Lean 4.

1 The conjecture

The statement under consideration, verbatim from the corpus, is the following.

Definition: A hypertree of the r -uniform complete hypergraph $K_n^{(r)}$ is a minimally connected subhypergraph of its edge set; $t(K_n^{(r)})$ denotes their count. Conjecture: For $r = 3$, $t(K_n^{(3)}) = n^{\binom{n-1}{2}-1} \cdot \prod_{i=1}^{n-1} f(i)$ with the explicit polynomial $f(i) = i^2 - i + 1$; this formula covers all known values for $n \leq 6$.

Write $V = \{1, \dots, n\}$ and let E_n be the set of all $\binom{n}{3}$ three-element subsets of V , so that $K_n^{(3)} = (V, E_n)$. A *subhypergraph* of $K_n^{(3)}$, in the only sense consistent with the phrase “subhypergraph of its edge set”, is a pair (V, F) with $F \subseteq E_n$; it is determined by F . A subhypergraph is *connected* if its edges cover V and the graph obtained by replacing each edge by a triangle on its three vertices is connected; it is *minimally connected* if it is connected and deleting any single edge destroys connectedness. A *hypertree* is a minimally connected subhypergraph, and $t(K_n^{(3)})$ counts them.

We write

$$\Phi(n) = n^{\binom{n-1}{2}-1} \prod_{i=1}^{n-1} (i^2 - i + 1)$$

for the conjectured value.

2 Refutation I: the counting bound

This is the load-bearing argument, and it is degenerate in its simplicity.

Lemma 1. *For every $n \geq 3$, $t(K_n^{(3)}) \leq 2^{\binom{n}{3}}$.*

Proof. A hypertree is a subhypergraph of $K_n^{(3)}$, hence is determined by a subset $F \subseteq E_n$ of its edges. There are $2^{|E_n|} = 2^{\binom{n}{3}}$ such subsets. The map $F \mapsto (V, F)$ is injective on subhypergraphs, so there are at most $2^{\binom{n}{3}}$ of them, and *a fortiori* at most that many hypertrees. $\square$

Lemma 1 does not mention connectedness, minimality, $r = 3$ beyond the edge count, or anything about the structure of hypergraphs. It survives every possible reading of the two undefined-looking words in the conjecture.

Theorem 1. $\Phi(n) > 2^{\binom{n}{3}}$ for every n with $3 \leq n \leq 13$. In particular the conjecture fails at $n = 3$.

Proof. For $n = 3$ we have $\binom{n-1}{2} - 1 = \binom{2}{2} - 1 = 0$ and $\prod_{i=1}^2 (i^2 - i + 1) = 1 \cdot 3 = 3$, so $\Phi(3) = 3^0 \cdot 3 = 3$, while $2^{\binom{3}{3}} = 2^1 = 2$. Thus $\Phi(3) = 3 > 2$, contradicting Lemma 1.

The remaining values are tabulated in Table 1; they are finite computations with integers. The bound is violated for $3 \leq n \leq 13$ and holds from $n = 14$ onwards, where $\binom{n}{3} \approx n^3/6$ finally outgrows $((\binom{n-1}{2} - 1) \log_2 n + \sum_{i < n} \log_2 (i^2 - i + 1)) \approx \frac{n^2}{2} \log_2 n$. $\square$

n	$\binom{n}{3}$	$2^{\binom{n}{3}}$	$\Phi(n)$	$\Phi(n)/2^{\binom{n}{3}}$
3	1	2	3	$1.5 \times$
4	4	16	336	$21 \times$
5	10	1024	853125	$833 \times$
6	20	1048576	57775431168	$55099 \times$
11	165	$\approx 5.0 \cdot 10^{49}$	$\approx 2.0 \cdot 10^{58}$	$\approx 4 \cdot 10^8 \times$
12	220	$\approx 1.6 \cdot 10^{66}$	$\approx 6.3 \cdot 10^{72}$	$\approx 4 \cdot 10^6 \times$
13	286	$\approx 1.3 \cdot 10^{86}$	$\approx 7.9 \cdot 10^{88}$	$\approx 8 \cdot 10^2 \times$
14	364	$\approx 4.0 \cdot 10^{109}$	$\approx 1.3 \cdot 10^{107}$	bound holds

Table 1: The conjectured formula against the number of subhypergraphs. The conjecture asserts agreement for $n \leq 6$; every one of those values is in the left-hand part of the table.

Remark 1. One counterexample suffices to refute a universal claim, and $n = 3$ is the smallest case in which $K_n^{(3)}$ is nonempty at all. It is worth noting how far the failure extends: the formula is not merely wrong at one value, it is impossible throughout the range the conjecture explicitly claims to have checked.

3 Refutation II: exact enumeration

The counting bound shows the formula is too large. Enumerating directly shows what the true values are.

$n = 3$. Here $|E_3| = \binom{3}{2} = 3$; the unique edge is $\{1, 2, 3\}$. There are two subhypergraphs, the empty one and the whole one. The empty one is not connected (its edges do not cover V), and the whole one is connected and vacuously minimal, since deleting the only edge disconnects it. Hence

$$t(K_3^{(3)}) = 1, \quad \Phi(3) = 3.$$

$n = 4$. Here $|E_4| = 6$. A single edge covers only three of the four vertices, so a connected subhypergraph needs at least two edges; two distinct triples cover all four vertices exactly when they omit different vertices, giving $\binom{4}{2} = 6$ pairs, each of which is connected and minimal. No set of three or more edges is minimal. Hence

$$t(K_4^{(3)}) = 6, \quad \Phi(4) = 4^2 \cdot (1 \cdot 3 \cdot 7) = 336.$$

$n = 5$. Here $|E_5| = 10$ and there are $2^{10} = 1024$ subhypergraphs. Exhaustive search gives

$$t(K_5^{(3)}) = 25, \quad \Phi(5) = 5^5 \cdot (1 \cdot 3 \cdot 7 \cdot 13) = 853125.$$

The 25 split as $15 + 10$: fifteen consist of two edges meeting in a single vertex and covering all five vertices, and ten consist of three edges sharing a common pair, the third entries running over the remaining three vertices.

n	subhypergraphs	$t(K_n^{(3)})$	$\Phi(n)$
3	2	1	3
4	16	6	336
5	1024	25	853125

Table 2: Exact counts against the conjectured formula.

The value $n = 6$ has $2^{20} = 1048576$ subhypergraphs; we do not enumerate it, because Theorem 1 already disposes of it.

4 Why no re-reading of the definition rescues it

The words “connected” and “minimal” admit some variation, so it is worth isolating what each refutation depends on.

- Theorem 1 uses only that a hypertree is determined by a subset of the edge set. Any definition of “minimally connected subhypergraph” for which this is true — which is to say, any definition at all — is refuted. In particular it does not matter whether connectivity is read via the 2-section, via incidence-graph connectivity, or via any stronger notion, and it does not matter whether “minimal” means minimal by inclusion or of minimum cardinality.
- The enumeration at $n = 3$ uses only that the empty subhypergraph is not connected and that the full one is. It is insensitive to the same variations.

The only reading we know of that would evade Lemma 1 is one in which a subhypergraph may also delete vertices, so that t counts hypertrees spanning *any* subset of V rather than all of V . Under that reading the notation $t(K_n^{(r)})$ would be misleading (it would depend only on n through the choice of available vertices), and the count at $n = 3$ would be 1 for the spanning hypertree plus one for each singleton vertex, i.e. 4, which still disagrees with $\Phi(3) = 3$. We record this only for completeness; the standard reading, and the one consistent with the classical notation $t(K_n) = n^{n-2}$ for graphs, is the spanning one.

5 Verification

- `python3 reproduce.py` — standalone, standard library only. Recomputes Table 1 and Table 2 from scratch by enumerating all $2^{\binom{n}{3}}$ subhypergraphs for $n = 3, 4, 5$ and checking connectedness and minimality of each.
- `lean4/` — a core Lean 4 project (no Mathlib, no sorry). Run `lake build && lake env lean Check.lean`.

The formalisation contains

- `sublists_length`: the powerset of an m -element set has 2^m elements;
- `filter_length_le`: filtering does not lengthen a list, hence `hypertrees_le_subhypergraphs`: $t(K_n^{(3)}) \leq 2^{\binom{n}{3}}$;
- `refutation_bound_3`: $\Phi(3) > t(K_3^{(3)})$, obtained by combining the bound with the exact value $2^{\binom{3}{3}} = 2$;
- `refutation_bound_6`: $\Phi(6) > 2^{\binom{6}{3}}$, i.e. $57775431168 > 2^{20}$;
- `hypertrees3_length`, `hypertrees4_length`, `hypertrees5_length`: the exact counts 1, 6, 25, each proved by `rfl`, so that the kernel itself performs the exhaustive search;
- `refutation_exact_3/4/5`: $\Phi(n) \neq t(K_n^{(3)})$ for $n = 3, 4, 5$.

Axiom audit (`#print axioms`): `hypertrees3_length`, `hypertrees4_length`, `hypertrees5_length` and the three `refutation_exact_*` theorems *depend on no axioms at all*. The bound theorems depend on `propext` and `Quot.sound`, inherited from the standard treatment of lists.

6 Scope

We refute the closed form as stated. We make no claim about the correct value of $t(K_n^{(3)})$ beyond $n \leq 5$, and no claim about the analogous formula for $r \neq 3$. The three exact values 1, 6, 25 are reported because they pin down the small cases, not because they suggest a pattern; three terms are not enough to guess one.